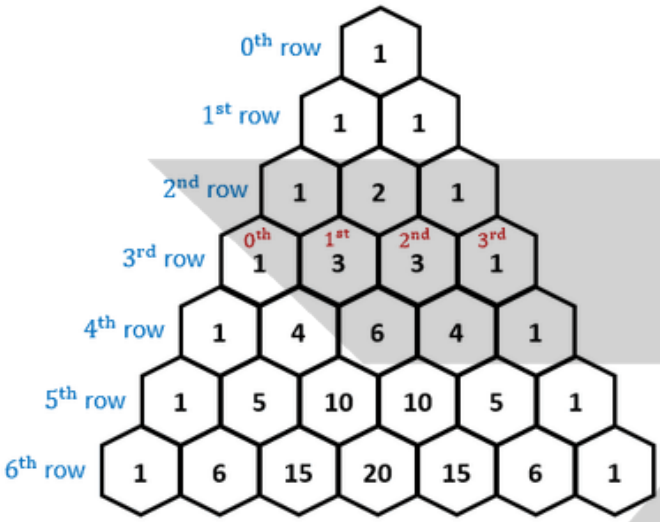
1. Series

1.1. Introduction to nCr notation

- Combinatorics helps us find the rth element in the nth row of pascal's triangle, which is a triangle that is constructed by summing the adjacent elements in the previous row.

Pascal's triangle



- As an example, from the definition, ${}^3C_1$ will give us a value of 3.
- The triangle is also symmetrical at the centre

Factorial notation

The factorial notation is used in the formula for combinations.

$$n! = n \times (n-1) \times (n-2) \times \cdots \times 2 \times 1$$

For example

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

It should also be known that

$$0! = 1$$

Combinations

The notation you should be familiar for combinations is

$${}^nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Using this, we obtain some useful results

$${}^nC_0 = {}^nC_n = 1$$

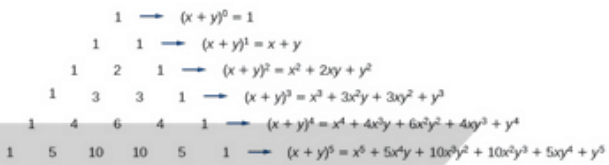
$${}^nC_1 = {}^nC_{n-1} = n$$

$${}^nC_r = {}^nC_{n-r}$$

- Reinstating, $\binom{n}{r}$ gives us the rth element in the nth row of pascal's triangle.
- Most calculators have a pre-defined function, or button, that allows you to use the ${}^nC_r$ formula directly.

1.2. Binomial expansion

- The binomial expansion helps us expand the sum of two terms raised to an exponent.
  - It makes use of pascal's triangle, as shown below, to obtain the binomial coefficient.



The expansion is given in MF19 for $n > 0$, and can be generalised to

$$(a+b)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \cdots + \binom{n}{n-1}ab^{n-1} + \binom{n}{n}b^n$$

- Observations:
  - The exponents for a and b sum up to the value of n
  - The $\binom{n}{r}$ notation can be thought of as **choosing** r number of terms to be the second term, b.

We can find the rth term of a binomial expansion using

$$T_{r+1} = {}^nC_r \times a^{n-r} \times b^r$$

Binomial expansion with 3 terms

- We can also use binomial expansion to expand a trinomial, or the sum of three terms raised to an exponent.

Consider the expansion of $(a+b)^n$

$$(a+b)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}b + \cdots + \binom{n}{n-1}ab^{n-1} + \binom{n}{n}b^n$$

We can write the trinomial as

$$(x+y+z)^n \implies (a+[y+z])^n$$

Now we can substitute $b = y+z$ in the expansion of $(a+b)^n$

$$(a+b)^n = \binom{n}{0}x^n + \binom{n}{1}x^{n-1}(y+z) + \cdots + \binom{n}{n-1}x(y+z)^{n-1} + \binom{n}{n}(y+z)^n$$

Example

Find the expansion of $(3x+2)^4$.

Answer

$$(3x+2)^4 = (3x)^4 + \binom{4}{1}(3x)^{4-1}(2) + \binom{4}{2}(3x)^{4-2}(2)^2 + \binom{4}{3}(3x)^{4-3}(2)^3 + \binom{4}{4}(2)^4$$

$$(3x+2)^4 = 81x^4 + (4)(3x)^3(2) + (6)(3x)^2(4) + (4)(3x)(8) + 16$$

$$(3x+2)^4 = 81x^4 + 216x^3 + 216x^2 + 96x + 16$$

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1.3. Arithmetic Progression

- An arithmetic progression (AP) is a series of numbers that share a common difference between consecutive terms
  - An example of an arithmetic progression is: $2, 4, 6, 8, \cdots$. Each consecutive term in this series has a common difference of 2.
  - The first term is denoted by a and the common difference is denoted by d

$$1^{st} \text{ term} = u_1 = a$$

$$2^{nd} \text{ term} = u_2 = a + d$$

$$3^{rd} \text{ term} = u_3 = a + 2d$$

$$\cdots$$

$$(n-1)^{th} \text{ term} = u_{n-1} = a + (n-2)d$$

$$n^{th} \text{ term} = u_n = a + (n-1)d$$

$$(n+1)^{th} \text{ term} = u_{n+1} = a + (n+1)d$$

- The formula for the nth term is also given in the MF19 formula sheet.

Finding the common difference

As mentioned previously, the common ratio must be the same between consecutive terms.

This can be written as

$$d = u_2 - u_1 = u_3 - u_2 = \cdots = u_n - u_{n-1} = u_{n+1} - u_n$$

We can re-write this as

$$d = a + d - a = a + 2d - (a + d)$$

$$\cdots$$

$$d = a + (n-1)d - [a + (n-2)d] = a + (n+1)d - [a + (n-1)d]$$

- Note that the difference of two consecutive terms results in d being the only term left.

Sum of Arithmetic Progression

- The sum of a geometric progression is the sum of terms in a given sequence.
  - S_n denotes the sum till the nth term.

It is given in MF19, and is denoted by

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

An alternate formula also exists

$$S_n = \frac{n}{2}(a + l)$$

Where l denotes the last term in the series

- The use of both formulae depends on the context, and information about the variables.

Example 1

Find the 120th term of the arithmetic progression: $2, 4, 6, 8, 10 \ldots$

Answer

- From the question
  - The first term is equal to a, which is 2 in this question.
  - The common difference, d, is equal to the difference between consecutive terms, $4 - 2 = 2$.

So to find the 120th term

$$u_{120} = 2 + (120 - 1)2 = 240$$

Example 2

Find the sum of the first 20 terms of the arithmetic progression, which has its first term equal to 7 and its 8th term being 28.

Answer

From the question, we can deduce that $a = 7$ as it is the first term. Using the general formula for the nth term for an AP

$$u_8 = 28 = 7 + (8-1)d$$

$$d = \frac{28 - 7}{7} = 3$$

Thus, we can use the sum till the nth to find the sum of the first 20 terms.

$$S_{20} = \frac{20}{2}[2(7) + (20-1)3] = 710$$

1.4. Geometric Progression

- A geometric progression (GP) is a series of numbers that share a common ratio between consecutive terms.
  - An example of a geometric sequence is: $3, 9, 27, 81, \cdots$. Each term in this series is being multiplied by a common ratio of 3.
  - The **first term** is given by a, while the **common ratio** is denoted by r

$$1^{st} \text{ term} = u_1 = a$$

$$2^{nd} \text{ term} = u_2 = ar$$

$$3^{rd} \text{ term} = u_3 = ar^2$$

$$\cdots$$

$$(n-1)^{th} \text{ term} = u_{n-1} = ar^{n-2}$$

$$n^{th} \text{ term} = u_n = ar^{n-1}$$

$$(n+1)^{th} \text{ term} = u_{n+1} = ar^n$$

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- The formula for the nth term is also given in the MF19 formula sheet

Finding the common ratio

As mentioned previously, the common ratio must be the same between consecutive terms.

This can be written as

$$r = \frac{u_2}{u_1} = \frac{u_3}{u_2} = \cdots = \frac{u_n}{u_{n-1}} = \frac{u_{n+1}}{u_n}$$

We can re-write this ratio as

$$r = \frac{ar}{a} = \frac{ar^2}{ar} = \cdots = \frac{ar^{n-1}}{ar^{n-2}} = \frac{ar^n}{ar^{n-1}}$$

- Note that the denominator cancels all the terms of the numerator, and leaves r as the only term.

Sum of Geometric Progression

- The sum of a geometric progression is the sum of terms in a given sequence.
  - S_n denotes the sum till the nth term.

It is given in MF19, and is denoted by

$$S_n = \frac{a(1-r^n)}{1-r} \quad [r < 1]$$

Another formula exists for the case r > 1

$$S_n = \frac{a(r^n-1)}{r-1} \quad [r > 1]$$

- The **sum till infinity** of a geometric progression, given in MF19, is denoted by

$$S_\infty = \frac{a}{1-r} \quad [-1 < r < 1]$$

- The inequality, $-1 < r < 1$, is also written as $|r| < 1$ but implies the same.
  - For a geometric progression to **converge**, or reach a finite value when the sum is taken till infinity, it **must satisfy** this inequality.

Example 1

Find the 10th term of the geometric progression, with the first term 3 and a ratio of $\frac{3}{4}$.

Answer

- From the question:
  - $a = 3$ and $r = \frac{3}{4}$, giving us $u_n = 3\left(\frac{3}{4}\right)^{n-1}$

So to find the 10th term,

$$u_{10} = 3\left(\frac{3}{4}\right)^{10-1} \implies 0.225$$

Example 2

The first 3 terms of a geometric progression are: $x+2$, $3x$, $\frac{9}{4}x^2$,where $x > 0$, show that $9x^3 - 18x^2 = 0$ and find the value of r and x.

Answer

We know the common ratio, r, is the ratio of any two consecutive terms

$$r = \frac{3x}{x+2} = \frac{\frac{9}{4}x^2}{3x}$$

Now we can begin simplifying our ratio

$$\frac{9x^2}{x+2} = \frac{9}{4}x^2$$

$$4(9x^2) = 9x^2(x+2)$$

$$36x^2 = 9x^3 + 18x^2$$

$$\implies 9x^3 - 18x^2 = 0$$

We can now find the value of r and x

$$9x^3 = 18x^2$$

$$\frac{x^3}{x^2} = \frac{18}{9} \implies x = 2$$

Substituting the value of x back into the common ratio gives us:

$$r = \frac{3(2)}{2+2} = \frac{3}{2}$$

1.5. Sum of a series

As introduced previously, the sum of a series adds up each term in a sequence till the nth term.

Using the general definition of the sum, we obtain

$$S_{n-1} = u_1 + u_2 + \cdots + u_{n-3} + u_{n-2}$$

$$S_n = u_1 + u_2 + \cdots + u_{n-2} + u_{n-1}$$

$$S_{n+1} = u_1 + u_2 + \cdots + u_{n-1} + u_n$$

Using these basic definitions we can derive two important results

$$S_n - S_{n-1} = u_{n-1}$$

$$S_{n+1} - S_n = u_n$$

- This can help you determine the common difference or ratio of a sequence

It is also important to note that:

$$S_1 = a \quad [a \text{ denotes the first term of the sequence here}]$$

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Sum of a series given in quadratic form

- Many questions may give you the sum in terms of a quadratic

$$S_n = An^2 + Bn \text{ [Where A and B are constants]}$$

- A sum given in this form is **always** an **arithmetic progression**.

One of the methods to solve such questions is using the definitions and results that arise from the sum of a series.

- An alternate way is to **compare the coefficients** with the sum of an arithmetic progression

$$S_n = \frac{n}{2}[2a + (n-1)d] = An^2 + Bn$$

$$\frac{dn^2}{2} - \frac{dn}{2} + an = An^2 + Bn$$

Upon comparing coefficients, we get:

$$\frac{d}{2} = A \implies d = 2A$$

$$-\frac{d}{2} + a = B \implies a = B + A$$

- Where a is the first term of the arithmetic sequence and d is the common difference.

Finding sum in-between 2 terms

- Recall that the formula for the sum of a series gives you the sum **till** the nth term.

To find the sum in-between two terms, say a and b [where $b > a$], we can use the fact that S_b includes S_a to some extent.

- To take an **example**, the sum of numbers **between** the 20th and 100th term for a sequence is given by:

$$S_{100} = u_1 + u_2 + \cdots + u_{19} + u_{20} + \cdots + u_{99} + u_{100}$$

$$S_{19} = u_1 + u_2 + \cdots + u_{18} + u_{19}$$

We can clearly deduce that

$$S_{100} - S_{19} = u_{20} + u_{21} + \cdots u_{99} + u_{100}$$

Another way to think about this is

$$S_{100} = S_{19} + u_{20} + u_{21} + \cdots + u_{99} + u_{100}$$

So subtracting S_{19} from S_{100} gives us the sum between the 20th and 100th term.

1.6. Trigonometry in series

- Trigonometric functions can be included in series and sequences.
  - They make use of identities such as $tan(\theta) = \frac{sin(\theta)}{cos(\theta)}$ and $sin^2(x) + cos^2(x) = 1$
  - These series can also exist for a specific domain, such as $0 \leq x \leq \pi$, but questions will usually specify the value of x or θ to use.

Trigonometric Arithmetic Progression

Example 1

Given an arithmetic sequence with first term $sin^2(\theta)$, where $0 < \theta < \frac{\pi}{2}$, and second term $sin^2(\theta)cos^2(\theta)$, find the common difference in terms of $sin(\theta)$

Answer

- From the question
  - $a = sin^2(\theta)$ and $a + d = sin^2(\theta)cos^2(\theta)$

$$\text{Using the fact that } u_{n-1} - u_n = d$$

$$sin^2(\theta)cos^2(\theta) - sin^2(\theta) \implies sin^2(\theta)(cos^2(\theta) - 1)$$

$$\text{After noticing that } cos^2(\theta) - 1 = -sin^2(\theta)$$

$$sin^2(\theta)(-sin^2(\theta)) \implies d = -sin^4(\theta)$$

Example 2

Given an arithmetic sequence with the nth term $u_n = sin^2(\theta) + (n-1)(-sin^4(\theta))$, find the sum of the first 16 terms when $\theta = \frac{\pi}{3}$.

Answer

Using $\theta = \frac{\pi}{3}$, we get

$$a = sin^2\left(\frac{\pi}{3}\right) = \frac{3}{4}$$

$$d = -sin^4\left(\frac{\pi}{3}\right) = -\frac{9}{16}$$

Now we can use the formula for the sum till the nth term

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

$$S_{16} = \frac{16}{2}[2\left(\frac{3}{4}\right) + (16-1)\left(-\frac{9}{16}\right)] = -\frac{111}{2}$$

Trigonometric Geometric Progression

Example

Given a geometric progression with first term $sin^2(\theta)$, where $0 < \theta < \frac{\pi}{2}$, and second term $sin^2(\theta)cos^2(\theta)$, find the sum till infinity.

Answer

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- From the question
  - We know that $a = sin^2(\theta)$ and $ar = sin^2(\theta)cos^2(\theta)$

To find the common ratio r:

$$r = \frac{sin^2(\theta)cos^2(\theta)}{sin^2(\theta)} = cos^2(\theta)$$

We can now use the sum till infinity formula for a geometric progression

$$S_\infty = \frac{sin^2(\theta)}{1 - cos^2(\theta)}$$

After noticing that $1 - cos^2(\theta) = sin^2(\theta)$

$$S_\infty = \frac{sin^2(\theta)}{sin^2(\theta)} = 1$$

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